

SIMATS ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – SOLUTION

Annexure A: Resolution Algorithm Implementation | Course: Artificial Intelligence (CSA17)

Each solution follows the standard five-stage resolution-refutation method: (1) formalize the knowledge base in propositional logic, (2) convert every implication to Conjunctive Normal Form (CNF) using $P \rightarrow Q \equiv (\neg P \vee Q)$, (3) negate the goal and add it to the clause set, (4) repeatedly resolve pairs of clauses containing complementary literals, and (5) stop when the Empty Clause (denoted NIL) is derived — a contradiction that proves the goal is entailed by the KB.

Question 1 – Rain and Wet Ground

Step 1: Propositional Logic Representation

Let **P** = 'It rains' and **Q** = 'The ground is wet'.

Premise 1: $P \rightarrow Q$ ("If it rains, then the ground becomes wet.")

Premise 2: **P** ("It is raining.")

Step 2: Conversion to CNF

$P \rightarrow Q \equiv \neg P \vee Q$ (implication elimination: $A \rightarrow B \equiv \neg A \vee B$)

C1: $(\neg P \vee Q)$

C2: **(P)**

Step 3: Negate the Goal

Goal: Prove that the ground is wet. $\rightarrow Q$. Negated goal: $\neg Q$

C3: $(\neg Q)$

Step 4: Apply the Resolution Algorithm

Step	Clause 1	Clause 2	Resolvent	Reason
1	C1: $(\neg P \vee Q)$	C2: (P)	C4: (Q)	Resolve on P / $\neg P$
2	C4: (Q)	C3: $(\neg Q)$	NIL (Empty Clause)	Resolve on Q / $\neg Q$

Step 5: Conclusion

Since the Empty Clause (NIL) was derived, the clause set is unsatisfiable, which proves by refutation that the original goal **Q** (Prove that the ground is wet) logically follows from the knowledge base.

✓ **GOAL PROVED — Empty Clause (NIL) derived in 2 resolution steps.**

Question 2 – Student Assignment Submission

Step 1: Propositional Logic Representation

Let **P** = 'A student submits the assignment' and **Q** = 'The student receives internal marks'.

Premise 1: $P \rightarrow Q$ ("If a student submits the assignment, then the student receives internal marks.")

Premise 2: **P** ("Rahul submitted the assignment.")

Step 2: Conversion to CNF

$P \rightarrow Q \equiv \neg P \vee Q$ (implication elimination: $A \rightarrow B \equiv \neg A \vee B$)

C1: $(\neg P \vee Q)$

C2: (P)

Step 3: Negate the Goal

Goal: Prove that Rahul receives internal marks. $\rightarrow Q$. Negated goal: $\neg Q$

C3: $(\neg Q)$

Step 4: Apply the Resolution Algorithm

Step	Clause 1	Clause 2	Resolvent	Reason
1	C1: $(\neg P \vee Q)$	C2: (P)	C4: (Q)	Resolve on $P / \neg P$
2	C4: (Q)	C3: $(\neg Q)$	NIL (Empty Clause)	Resolve on $Q / \neg Q$

Step 5: Conclusion

Since the Empty Clause (NIL) was derived, the clause set is unsatisfiable, which proves by refutation that the original goal **Q** (Prove that Rahul receives internal marks) logically follows from the knowledge base.

✓ **GOAL PROVED — Empty Clause (NIL) derived in 2 resolution steps.**

Question 3 – Library Membership

Step 1: Propositional Logic Representation

Let **P** = 'A person is a library member' and **Q** = 'The person can borrow books'.

Premise 1: $P \rightarrow Q$ ("If a person is a library member, then the person can borrow books.")

Premise 2: P ("Priya is a library member.")

Step 2: Conversion to CNF

$P \rightarrow Q \equiv \neg P \vee Q$ (implication elimination: $A \rightarrow B \equiv \neg A \vee B$)

C1: $(\neg P \vee Q)$

C2: (P)

Step 3: Negate the Goal

Goal: Prove that Priya can borrow books. $\rightarrow Q$. Negated goal: $\neg Q$

C3: $(\neg Q)$

Step 4: Apply the Resolution Algorithm

Step	Clause 1	Clause 2	Resolvent	Reason
1	C1: $(\neg P \vee Q)$	C2: (P)	C4: (Q)	Resolve on $P / \neg P$
2	C4: (Q)	C3: $(\neg Q)$	NIL (Empty Clause)	Resolve on $Q / \neg Q$

Step 5: Conclusion

Since the Empty Clause (NIL) was derived, the clause set is unsatisfiable, which proves by refutation that the original goal **Q** (Prove that Priya can borrow books) logically follows from the knowledge base.

✓ **GOAL PROVED — Empty Clause (NIL) derived in 2 resolution steps.**

Question 4 – Placement Eligibility

Step 1: Propositional Logic Representation

Let **P** = 'A student clears the aptitude test' and **Q** = 'The student is eligible for placement'.

Premise 1: $P \rightarrow Q$ ("If a student clears the aptitude test, then the student is eligible for placement.")

Premise 2: **P** ("Arun cleared the aptitude test.")

Step 2: CNF Clauses

C1: $(\neg P \vee Q)$

C2: **(P)**

Step 3: Negate the Goal

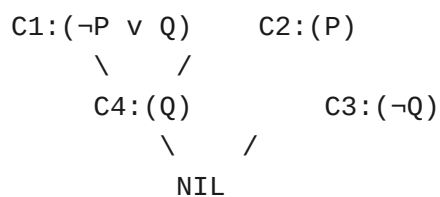
Goal: **Q** ('Arun is eligible for placement'). Negated goal: $\neg Q$

C3: $(\neg Q)$

Step 4: Resolution Steps

Step	Clause 1	Clause 2	Resolvent	Reason
1	C1: $(\neg P \vee Q)$	C2: (P)	C4: (Q)	Resolve on P / $\neg P$
2	C4: (Q)	C3: $(\neg Q)$	NIL (Empty Clause)	Resolve on Q / $\neg Q$

Step 5: Resolution Tree



Conclusion

The derivation of the Empty Clause (NIL) proves by refutation that **Q** ('Arun is eligible for placement') is a logical consequence of the knowledge base.

✓ **GOAL PROVED — Empty Clause (NIL) derived in 2 resolution steps.**

Question 5 – Access Control System

Step 1: Propositional Logic Representation

Let **P** = 'User enters the correct password', **Q** = 'User is authenticated', **R** = 'User is granted access'.

Premise 1: $P \rightarrow Q$ (correct password $\Rightarrow$ authenticated)

Premise 2: $Q \rightarrow R$ (authenticated $\Rightarrow$ granted access)

Premise 3: **P** (user entered the correct password)

Step 2: Conversion to CNF

C1: $(\neg P \vee Q)$

C2: $(\neg Q \vee R)$

C3: **(P)**

Step 3: Negate the Goal

Goal: R ('the user is granted access'). Negated goal: $\neg R$
C4: ($\neg R$)

Step 4: Apply the Resolution Algorithm

Step	Clause 1	Clause 2	Resolvent	Reason
1	C1: ($\neg P \vee Q$)	C3: (P)	C5: (Q)	Resolve on P / $\neg P$
2	C5: (Q)	C2: ($\neg Q \vee R$)	C6: (R)	Resolve on Q / $\neg Q$
3	C6: (R)	C4: ($\neg R$)	NIL (Empty Clause)	Resolve on R / $\neg R$

Explanation of Each Step

Step 1 chains the password premise (C3) with the authentication rule (C1) to derive **Q** — the user is authenticated. **Step 2** chains that result with the access-granting rule (C2) to derive **R** — the user is granted access. **Step 3** resolves this derived fact (R) against the negated goal ($\neg R$), producing a direct contradiction — the Empty Clause (NIL). This demonstrates two-step forward chaining collapsed into a resolution refutation.

Step 5: Final Conclusion

Because assuming $\neg R$ (access NOT granted) together with the knowledge base leads to a contradiction (NIL), the original goal **R** must be TRUE. Hence it is proved that the user is granted access.

✓ GOAL PROVED — Empty Clause (NIL) derived in 3 resolution steps.